

Total No. of Questions : 8]

SEAT No. :

PC2818

[6352]-42

[Total No. of Pages : 2

**S.E. (Computer Engineering) (Computer Science & Design Engg.)
(Artificial Intelligence & Data Science Engg.)**

**OBJECT ORIENTED PROGRAMMING
(2019 Pattern) (Semester-III) (210243)**

Time : 2 ½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Draw Neat and Clean Diagram
- 3) Assume Suitable Data if Necessary

- Q1)** a) Explain what is type conversions? Define implicit type casting and explicit type casting with example. [5]
- b) Explain Virtual destructor with the help of a program. [6]
- c) Write a C++ program to overload binary operator '+' to add two complex numbers using member function and friend function. [6]

OR

- Q2)** a) Explain Polymorphism and Types of Polymorphism. [5]
- b) How do you declare and define a pure virtual function in C++?
Explain with help of a program. [6]
- c) Explain need of operator overloading. Write C++ program to demonstrate use of unary operator overloading. [6]

- Q3)** a) Explain overloading the extraction (>>) and insertion (<<) operators with the help of a program. [5]
- b) What are File Pointers? Explain seekg() and tellg() functions with syntax [6]
- c) Write a program that emulates the DOS COPY command. That is, it should copy the contents of a text file (such as any .CPP file) to another file. Invoke the program with two command-line arguments-the source file and the destination file-like this; C>ocopy srcfile.cpp destfile.cpp In the program, check that the user has typed the correct number of command-line arguments, and that the files specified can be opened. [7]

OR

P.T.O.

- Q4)** a) List and Explain different Mode bits used in open() function, while opening a file. (Any five) [5]
- b) Define a class Book that has three attributes viz bookname, author and price. Write a C++ Program that writes an object to a file and reads an object from a file. [6]
- c) What do you mean by File Handling? Explain following functions with syntax and an example. [7]
- i) open()
 - ii) read()
 - iii) write()

- Q5)** a) Explain fundamentals of exception in C++ with suitable example. [5]
- b) Discuss class template with suitable example in C++. [6]
- c) Write a program in C++ to demonstrate Divide by Zero exception. [6]

OR

- Q6)** a) Write a program to demonstrate user defined exception in C++. [5]
- b) What is template in C++? Discuss template with multitype parameter in C++. [6]
- c) Write a program to demonstrate overloading function template in C++. [6]

- Q7)** a) Write a program to demonstrate a map in C++. [5]
- b) Enlist various containers in STL. Explain any one with the help of a program. [6]
- c) What is STL? Enlist and explain various components of STL. [7]

OR

- Q8)** a) What is an iterator? What are the various types of iterators? [5]
- b) What is Map? Enlist and explain various functions of map. [6]
- c) What is a vector? Write a C++ program to explain various functions of vector. [7]

